Evaluate a Functional-Type Classification Change

Description

Compares two functional-type classifications by taxon, raw cluster labels, label-invariant co-clustering relationships, and silhouette widths. The partition comparison is invariant to arbitrary integer relabelling of otherwise identical clusters.

Usage

```
evaluate_functional_type_classification_change(data_reference, data_candidate)
```

Arguments

`data_reference`	A data frame with unique 'taxon_name' values and columns 'functional_type' and 'silhouette_width'.
`data_candidate`	A data frame with the same required columns as 'data_reference'.

Details

Label-invariant partition differences are counted from cluster-size and contingency-table combinations. This avoids constructing a quadratic taxon-by-taxon matrix while counting pairs that are in the same cluster in exactly one classification.

Value

A named list with 'data_summary', a one-row tibble of comparison counts, and 'data_taxon_comparison', a taxon-level full join of both classifications.

Examples

```
data_reference <- tibble::tibble(
  taxon_name = c("a", "b", "c"),
  functional_type = c(1L, 1L, 2L),
  silhouette_width = c(0.4, 0.5, 0.6)
)
data_candidate <- tibble::tibble(
  taxon_name = c("a", "b", "c"),
```

```
  functional_type = c(2L, 2L, 1L),
  silhouette_width = c(0.4, 0.5, 0.6)
)
evaluate_functional_type_classification_change(
  data_reference,
  data_candidate
)
```